

Binomial Theorem

1st term 2nd term

$$(X + y)^2 = X^2 + 2xy + y^2 \quad \text{number of terms} = n+1$$

$$(ax + by)^n = {}^nC_0 (by)^0 (ax)^{n-0} + {}^nC_1 (by)^1 (ax)^{n-1} \\ + {}^nC_2 (by)^2 (ax)^{n-2} + \dots + {}^nC_n (by)^n$$

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$$n! = n(n-1)(n-2) \dots (3 \times 2 \times 1)$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

General Term

$$\left[T_{r+1} = {}^nC_r (\text{2nd})^r (\text{1st})^{n-r} \quad \frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \right]$$

Ex. Find the term contain X^4 in the expansion $(\frac{2}{x} + 3x^2)^5$

$$T_{r+1} = {}^5C_r (3x^2)^r (\frac{2}{x})^{5-r} \quad \text{Soln}$$

$$(3x^2)^r (\frac{2}{x})^{5-r} = (3)^r \cdot (2)^{5-r} \cdot X^{2r} \cdot X^{r-5} = (3)^r (2)^{5-r} (X)^{3r-5}$$

$$3r-5=4 \rightarrow 3r=9 \rightarrow r=3 \quad \therefore \text{Term is } T_4$$

Ex. If $\frac{T_7}{T_6} = 1$ in the expansion $(2x+3y)^n$ Then find n

Soln

$$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \rightarrow \frac{T_7}{T_6} = \frac{n-6+1}{6} = \frac{n-5}{6} = 1 \rightarrow n-5=6 \\ \Rightarrow n=11$$

Ex. In the expansion $(2x-3y)^{10}$ find the greatest term's value at $x=3$ and $y=2$.

Soln Suppose that T_{r+1} is the greatest term

$$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} = \frac{10-r+1}{r} = \frac{11-r}{r} > 1$$

$$\hookrightarrow 11-r > r \rightarrow 11 > 2r \Rightarrow r < \frac{11}{2} \Rightarrow r < 5.5$$

$$\frac{T_{r+2}}{T_{r+1}} = \frac{n-(r+1)+1}{r+1} = \frac{n-r}{r+1} = \frac{10-r}{r+1} < 1 \Rightarrow 10-r < r+1$$

$$\hookrightarrow 2r > 9 \Rightarrow r > \frac{9}{2} \Rightarrow r = 5 \text{ greatest term is } T_6$$

$$T_6 = {}^{10}C_5 (-3y)^5 (2x)^{10-5} = 252 \cdot (-243) \cdot 1024 = -622080$$